

Adaptive AutoSAR Graduation Project

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Automotive Mechatronics Engineering Department



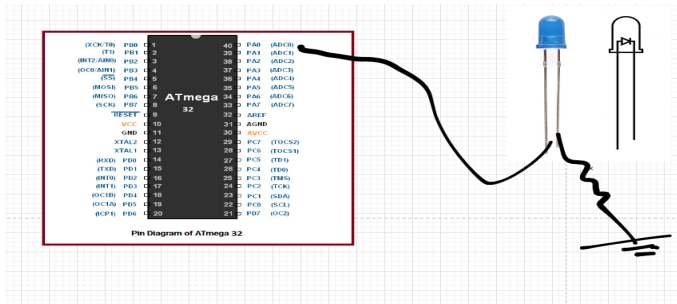
- 1 Classic Autosar
- 2 Automotive Industry Transformation
- 3 Impact on MCU Design
- 4 Exploring AP
- 5 Conclusion



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Introduction



- NonAutoSAR Application: Toggle the led on pin A0 each 1 sec.

Code

```
#include <avr/io.h>
#include <util/delay.h>

int main(void) {
    // Set PB0 as an output pin by setting its corresponding bit in DDRB to 1
    DDRB |= 0b00000001;

    while (1) {
        // Turn on the LED by setting PB0 high (assigning 1 to PB0)
        PORTB |= 0b00000001;
        _delay_ms(1000); // Wait for 1 second

        // Turn off the LED by setting PB0 low (assigning 0 to PB0)
        PORTB &= 0b11111110;
        _delay_ms(1000); // Wait for 1 second
    }

    return 0;
}
```

- Datasheet:
 - 1- specify pin to be OUTPUT
 - 2- generative 5v to be OUTPUT HIGH

Imagine

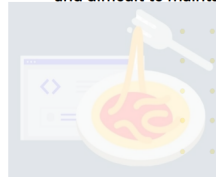
Imagine

What we will do to
convert the
microcontroller

Imagine

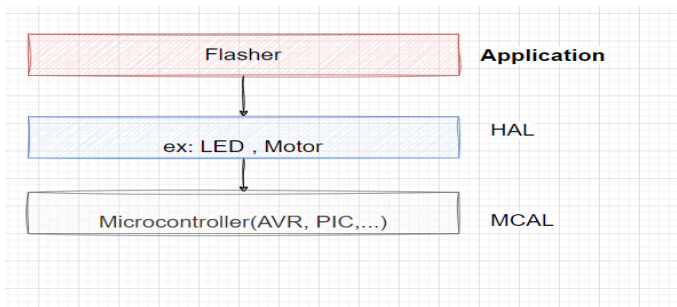
Having a source code
file contains 10
thousands lines of
code. If you have a
bug in this code, how
complex is finding the
bug

This is what's called the
Spaghetti Code, which is
the code that unstructured
and difficult to maintain

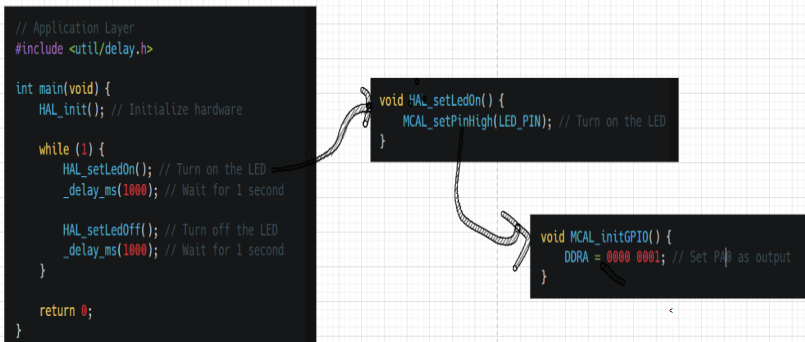


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Layered Architecture

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Layered Architecture



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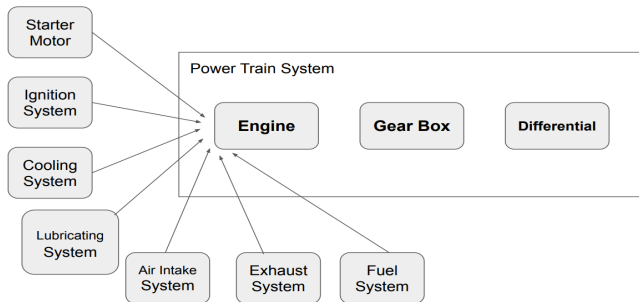


Today's Automotive System



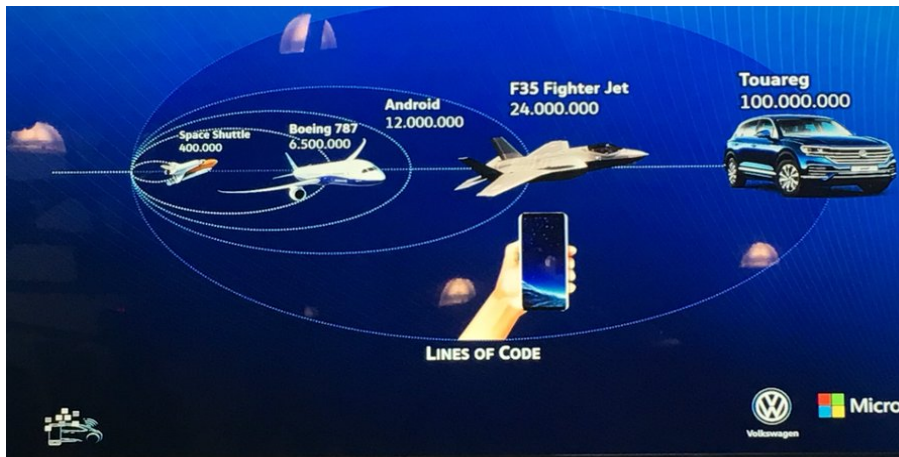
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PTS

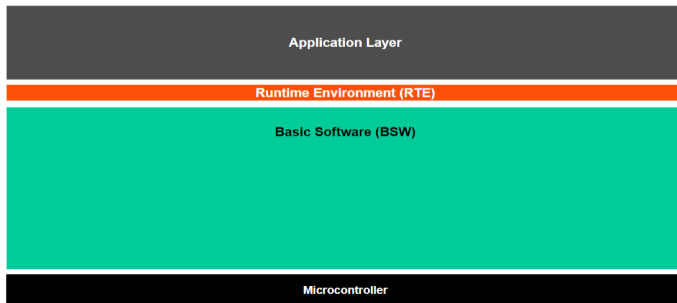


- Each System inside PTS System within other Automotive system has at least one ECU with its own layered Architecture.
- 100 million lines

Lines Of code



CP Architecture

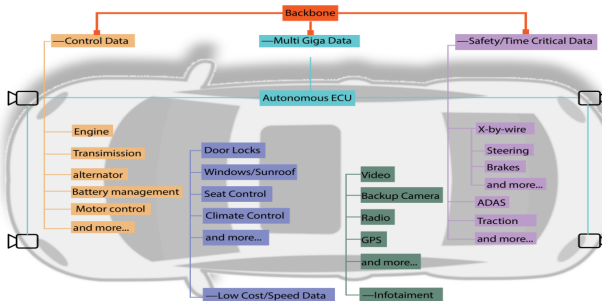
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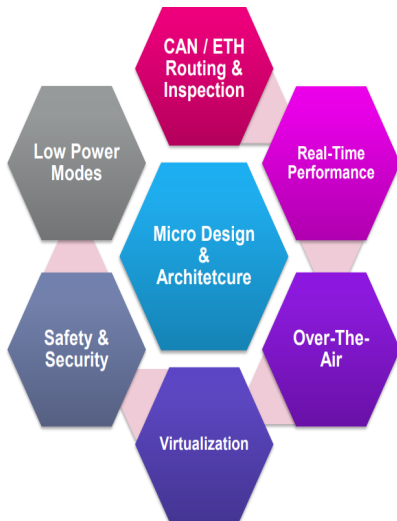


Vehicle Network Applications



- **Safety/Time Critical Data:** Requires relative high speed $> 10\text{Mb/s}$ with good time synchronization and redundancy.
- **Control Data:** General Purpose for transporting data from one system to another, generally packet data. Gateways
- **Backbone:** General Purpose for transporting data from one system to a another, generally packet data. Gateways
- **Infotainment:** streaming audio/video data critical. Requires good time-sync and control data, high speed $> 25\text{Mb/s}$
- **Multi Giga Data:** Currently very high Bandwidth for point to point networks specialized for uncompressed video in AI
- **Low cost/speed LOW:** As cheap as possible for slow speed non-safety control: door switches, lighting, HVAC control.

HW Requirements



CAN / ETH Routing & Inspection

- Routing performance
- Packet inspection
- Software overload
- New protocols

Real Time Behavior

- Start-up / power-down
- Reaction time
- Accelerators

Over The-Air

- Hardware support
- Rollback functionality
- External Memory interfaces
- Secure boot

Virtualization

- HW hypervisor
- Real-time, deterministic behavior

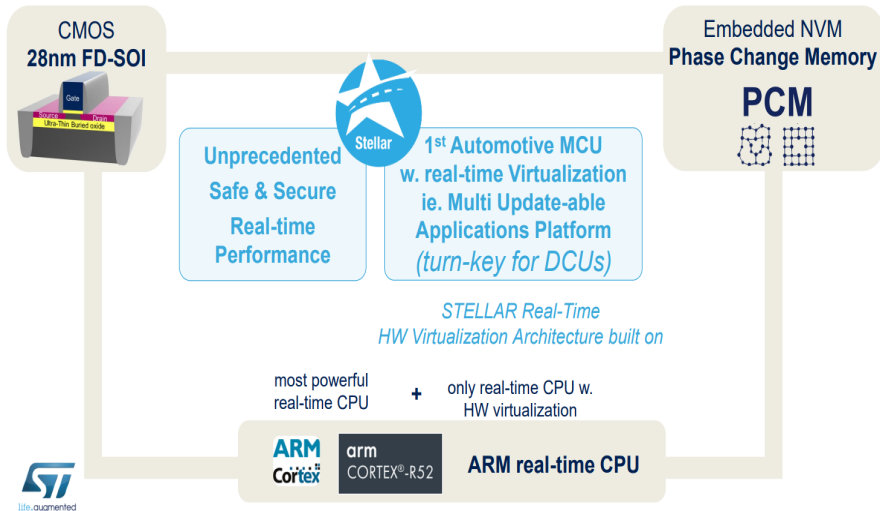
Safety & Security

- ASIL-D support
- HW self-test
- HSM subsystem
- Memory protection
- Communication security

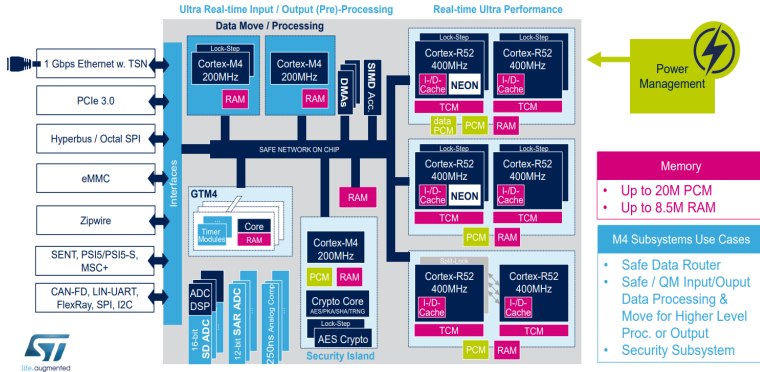
Low Power Modes

- Standby/Stop operations
- Smart wakeup unit

HW Requirements



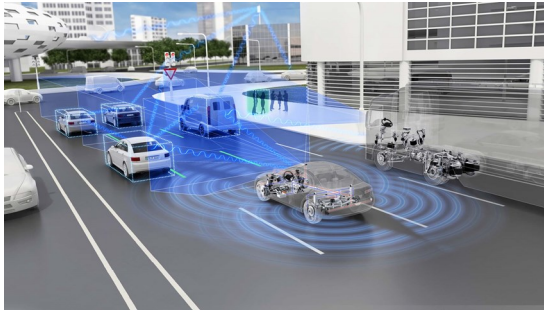
HW Capabilities



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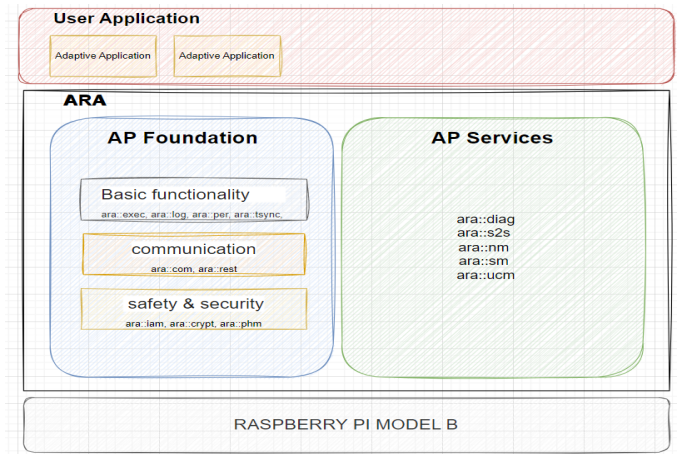
Adaptive Autosar AP



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New functionalities such as environment perception, autonomous driving, vehicle connectivity and Car as a Service(CaaS), many new functionalities and also many complexities have introduced a new set of problems to the electrical/electronic (E/E) architecture.

Architecture



Adaptive Will Not Replace classic

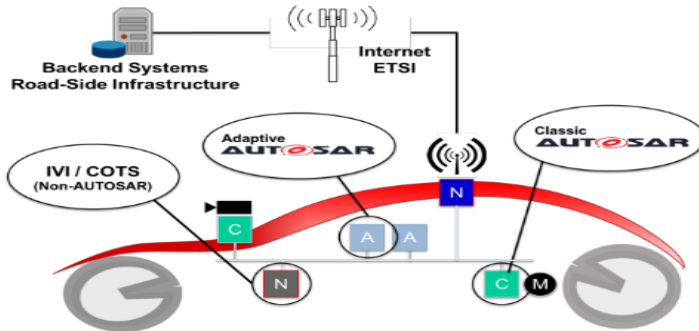


Figure 3.1: Exemplary deployment of different platforms

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Conclusion

- Adaptive AUTOSAR comes to life as new requirements and new resources come into the automotive industry. The new requirements, like autonomous driving, vehicle connectivity and constant updates, and the new resources, like high performance computing and ethernet communications, cannot be perfectly fulfilled by the existing AUTOSAR standard and therefore require a new rule set to guide them and set order in what otherwise would be another chaotic scenario. What the original AUTOSAR was to the deeply embedded ECUs, Adaptive AUTOSAR is to the new generation of vehicle computers.